

Engineering Report: Dynamic Sensor-Fusion Routing

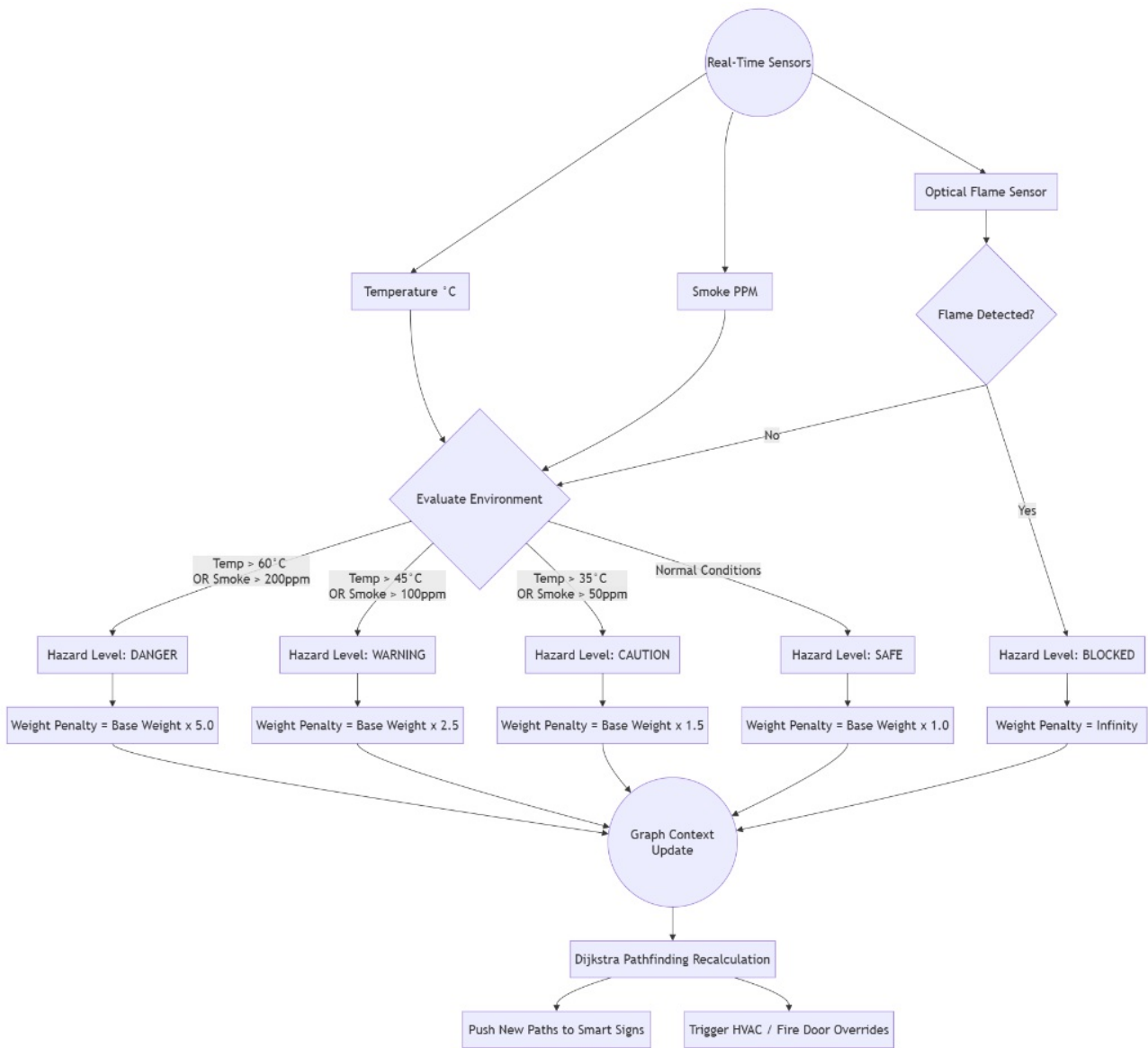
1. System Overview

Fire Commander utilizes a centralized graph-based network where hallways and rooms are **Edges** and intersections/exits are **Nodes**. To route occupants safely during an emergency, the system employs **Dijkstra's Algorithm** to find the path of least resistance (lowest total weight).

Instead of relying on static distances, the system fuses real-time telemetry (Temperature, Smoke PPM, and Flame Detection) to dynamically penalize paths that are becoming dangerous.

2. Sensor-Fusion Flowchart

The following flowchart illustrates the exact decision tree used to convert raw sensor data into actionable edge-weights for the pathfinding engine.



3. Mathematical Weighting Model

The traversal cost (\$C\$) of any given edge (\$E\$) is calculated dynamically using the base distance (\$D\$) and a hazard penalty multiplier (\$P_h\$).

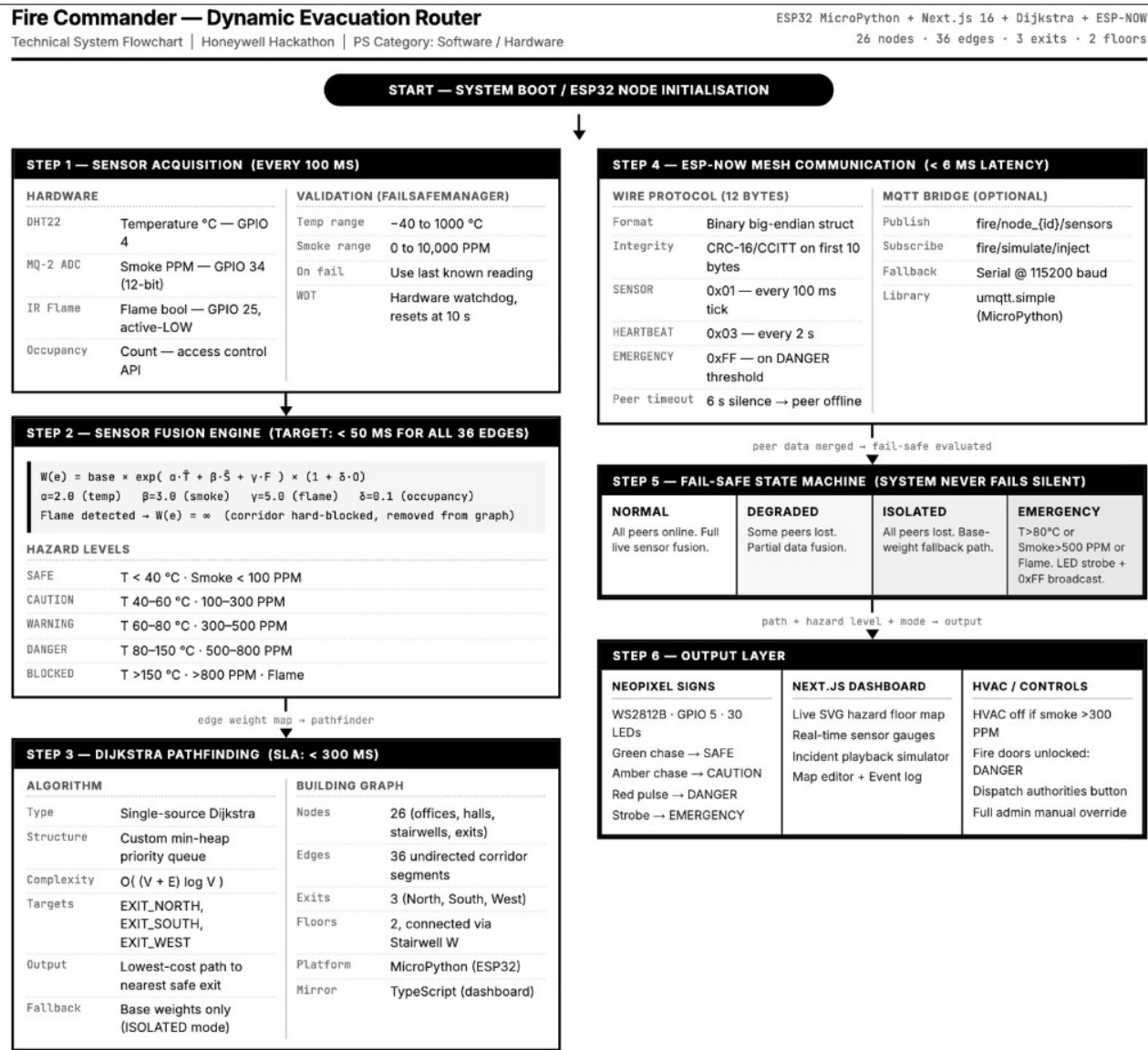
$$C_E = D_E \times P_h$$

Hazard Penalties (\$P_h\$)

- **SAFE (\$P_h = 1.0\$):** Normal walking speed. Occupants can traverse the area safely.
- **CAUTION (\$P_h = 1.5\$):** Slight smoke or heat. Path is viable but less optimal.
- **WARNING (\$P_h = 2.5\$):** Moderate smoke. Visibility is dropping. The algorithm will strongly prefer longer, clearer routes unless no other option exists.
- **DANGER (\$P_h = 5.0\$):** Heavy smoke or dangerous heat. The algorithm will route occupants through this edge *only* as an absolute last resort to escape.
- **BLOCKED (\$P_h = \infty\$):** Active flame detected. The edge is severed mathematically. The algorithm will drop this edge from the graph entirely.

4. Technical System Architecture

To fully grasp the scope of Fire Commander, we must look beyond the abstract mathematical model and examine the real-world hardware and software implementation. The system is designed as a highly distributed, fault-tolerant network of edge nodes communicating with a central command dashboard.



- **EMERGENCY:** If extreme conditions are met locally, the node automatically triggers an emergency strobe regardless of network connectivity.

Step 6: The Output Layer

The final computed paths and hazard states are pushed to three distinct output channels simultaneously:

1. **Physical Smart Signs:** WS2812B Neopixel strips at corridor intersections display dynamic green chasing arrows pointing to safety, amber warnings, or red emergency pulses.
2. **Next.js Command Dashboard:** Administrators receive a live, SVG-rendered map of the building, real-time sensor gauges, and incident logs.
3. **Automated HVAC & Door Controls:** The system physically interfaces with the building's infrastructure—automatically suppressing HVAC ventilation in smoke-filled zones (to starve fires of oxygen) and unlocking electronic fire doors to prevent entrapment.

Conclusion

Fire Commander represents a paradigm shift in emergency evacuation. By marrying low-latency edge hardware with an intelligent, centralized Next.js routing engine, it transforms passive building infrastructure into an active, life-saving guidance system.